

MANUELA, Antarctica

WMO: 898640

Lat: 74.946S

Lon: 163.687E

Elev: 80

StdP: 100.37

Time Zone: 12.00 (NZL)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-37.5	-36.1	N/A	N/A	N/A	N/A	N/A	N/A	43.2	-21.0	40.4	-24.5	7.2	290	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.6	2.6	N/A	1.4	N/A	0.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.2	27.6	22.9	DB	-40.5	5.9	2.9	1.7	-42.6	7.1	-44.3	8.1	-45.9	9.1	-48.0	10.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-18.5	-3.5	-10.8	-19.9	-24.6	-25.4	-25.1	-26.8	-27.0	-24.8	-19.7	-10.0	-3.6	(6)	
	DBStd	9.65	2.37	3.67	4.23	4.50	4.90	4.83	5.44	5.46	4.87	3.85	3.87	2.36	(7)	
	HDD10.0	10388	420	583	926	1038	1097	1053	1141	1146	1044	919	599	421	(8)	
	HDD18.3	13429	678	816	1184	1288	1356	1303	1399	1404	1294	1178	849	679	(9)	
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.7	5.0	7.2	11.9	9.4	9.1	9.7	8.7	7.6	6.0	8.6	5.6	3.6	(14)	
(15) Precipitation	PrecAvg	252	28	24	24	17	20	22	17	22	19	19	19	20	(15)	
	PrecMax	334	58	67	54	41	48	43	35	56	48	41	59	42	(16)	
	PrecMin	135	7	11	10	3	2	6	3	2	0	3	4	3	(17)	
	PrecStd	48	13	11	10	9	11	10	8	14	13	10	14	10	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	-1.0	-7.3	-8.9	-10.5	-10.2	-6.2	-8.3	-10.4	-7.8	0.2	4.4	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	2.6	-2.9	-10.1	-12.9	-14.0	-12.8	-13.1	-12.9	-12.6	-10.8	-2.3	2.8	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	1.5	-4.4	-12.2	-15.8	-16.3	-15.7	-17.0	-16.7	-15.2	-12.6	-3.5	1.8	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	0.4	-5.8	-14.1	-18.3	-18.5	-18.3	-19.5	-19.0	-17.6	-14.3	-4.6	0.7	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	5.5	4.9	4.9	5.1	5.4	5.2	5.6	5.6	6.2	6.7	6.8	(35)	
		MCDBR	7.8	6.9	6.1	7.4	6.4	7.3	7.7	7.6	7.2	7.4	7.5	8.0	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub	0.240	0.225	0.184	0.111	N/A	N/A	N/A	N/A	0.109	0.173	0.199	0.221	0.237	(40)	
	taud	2.303	2.299	2.242	1.904	N/A	N/A	N/A	N/A	1.882	2.191	2.277	2.284	2.293	(41)	
	Ebn at Noon	982	921	822	418	0	0	0	0	396	823	956	1003	1006	(42)	
	Edh at Noon	98	80	53	14	0	0	0	0	13	53	81	99	104	(43)	
(44) All-Sky Solar Radiation	RadAvg	8.41	5.09	2.08	0.30	0.00	0.00	0.00	0.00	0.08	1.26	4.10	7.78	9.77	(44)	
	RadStd	0.18	0.20	0.09	0.02	0.00	0.00	0.00	0.00	0.01	0.07	0.12	0.21	0.25	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page